1. Suppose S1 entails S2 and S2 is unsatisfiable. Then what can be said about S1:
    a. Satisfiable
    b. Valid
    c. Unsatisfiable
    d. Not enough information

    S1 ⇒ S2

S1	S2	S1 ⇒ S2
0	0	1
0	1	1
1	0	0
1	1	1

    Since we know S1 ⇒ S2 only rows 1,2 and 4 are possible. S2 is unsatisfiable means S1 has to be unsatisfiable (row 1).

2. Suppose S1 entails S2 and S1 is unsatisfiable. Then what can be said about S2:
    a. Satisfiable
    b. Valid
    c. Unsatisfiable
    d. Not enough information

    S1 ⇒ S2

S1	S2	S1 ⇒ S2
0	0	1
0	1	1
1	0	0
1	1	1

    Since we know S1 ⇒ S2 only rows 1,2 and 4 are possible. S1 is unsatisfiable means S2 can be either true or false (row 1,2). Therefore S2 is satisfiable.
    Wrong answer choice in the Quiz. Question to be deleted.

3. You are given 10 binary variables and a CNF clause containing 10 literals - one literal for each variable. How many value assignments to the 10 variables are possible which make this clause true?
   Ans. 1023

   Total number of variable assignments = 2^{10} = 1024
   CNF clause can be false only when all the variables are opposite to how they appear in the clause.
   e.g. For 3 variable case, if we have clause {x1,-x2,x3} then it can be false only when x1=0,x2=1,x3=0.
   Thus, number of satisfying variable assignments = 1023

4. GSAT is an example of which of the following:
   a. Hill-Climbing with random restarts
   b. Random Walk
   c. Hill Climbing with random walk
   d. Genetic Algorithm

   Covered in class slides.
   GSAT involves combination of greedy (hill climbing) and random moves(random walk)

5. Which of the following are advantages of DPLL over WalkSAT?
   a. WalkSAT cannot tell if the problem is UNSAT while DPLL can detect UNSAT problems
   b. WalkSAT is not sound but DPLL is sound
   c. None of the above

   DPLL can conclude that the problem is UNSAT after exhausting the search, whereas WalkSAT can not conclude so since it is a local search.

6. Which of the following statement(s) is/are correct about resolution in propositional logic?
   a. If S is a set of clauses that resolve to a clause R then S entails R.
   b. Under the resolution technique, P v P can be resolved to P.
   c. If S is a set of clauses that entail a clause R then S resolves to R.
   d. Resolution is a powerful technique to prove a set of clauses unsatisfiable.

   http://www.cse.iitd.ac.in/~mausam/courses/col333/autumn2019/lectures/07-logic.pdf
   a. Resolution is sound therefore if S resolves to R then S entails R. [32]
   b. Resolution cannot infer P from P v P (i.e. it is not complete). [34]
   c. Eg. Consider S = {P v P} and R = {P}. Clearly, S entails R but S does not resolve to R.
   d. Using refutation completeness of resolution [32]

7. Which of the following components of a Knowledge Representation is used to encode the grammar that represents all sentences in the language?
    a. Semantics
    b. Knowledge Base
    c. Syntax
    d. Inference Engine

    http://www.cse.iitd.ac.in/~mausam/courses/col333/autumn2019/lectures/07-logic.pdf
    The syntax is used to represent sentences in a language. [7]

8. Find the first assignment to the following CNF formula using model enumeration inference with MOM's heuristic. Resolve clashes in selection using lexicographic order of variables and in the assignment using true over false. Report the answer as in a consecutive string of values of abcde. For example, report the answer as 01110 if a is false, b is true, c is true, d is true, and e is false.

    { c v ¬d , ¬b v ¬c , a v ¬c v e , b v d v ¬e , ¬a v ¬d }

    Ans: 10100

    MOM's heuristic: Most occurrences in clauses of minimum length.

Clauses	Clauses of min length	Variable	Value
c v¬d, ¬bv¬c, av¬cve, bvdv¬e, ¬av¬d	cv¬d,¬bv¬c,¬av¬d	c	true
¬b, ave, bvdv¬e, ¬av¬d	¬b	b	false
ave, dv¬e, ¬av¬d	ave, dv¬e, ¬av¬d	a	true
dv¬e, ¬d	¬d	d	false
¬e	¬e	e	false

9. Consider the following CNF formula. If x is the number of unit literals and y is the number of pure literals then find the value of 10x + y.

    { ¬c , ¬b v c , c v ¬e , b v d v ¬e , ¬a v ¬d , b }

    Ans: 22

    b, ¬c are unit literals as they form a unit literal clause.
    ¬a, ¬e are pure literals because no clause contains a or e literals.
    x = 2, y = 2 therefore 10x + y = 22.

10. Which of the following statements is/are correct about WalkSAT?
    a. The number of clauses satisfied increases by at least one in every move.
    b. It is guaranteed to make one false clause true in every one move.
    c. It is asymptotically complete.
    d. It iterates over all variables to find the best one to tune.

WalkSAT satisfies one clause at every move but another clause may violate. Increasing the total no. of satisfied clauses by one at every move would mean that the WalkSAT algorithm is linear in the no. of clauses and is complete and optimal. WalkSAT is asymptotically complete and it just iterates over a subset of variables to find the best one to tune.